

ERP SEKOLAH ALAM

Enterprise Architecture Document

Level 1 – Level 6

Version 1.0

LEVEL 1 — BUSINESS DISCOVERY

1. Business Goal

Visi Sistem

Menyediakan platform terpadu yang mengelola seluruh siklus pendidikan Sekolah Alam mulai dari penerimaan siswa baru, kegiatan akademik, pengukuran Learning Outcome, pembelajaran digital, pengelolaan kegiatan sekolah, hingga pelaporan kepada orang tua.

Tujuan Bisnis

1. Digitalisasi proses PPDB.
 2. Sentralisasi data siswa, guru, dan orang tua.
 3. Mendukung pembelajaran berbasis alam dan pengalaman.
 4. Mendukung pengukuran Learning Outcome Assessment (LOA).
 5. Menyediakan komunikasi yang transparan kepada orang tua.
 6. Mempermudah pengelolaan event sekolah.
 7. Mengelola tagihan pendidikan secara terstruktur.
 8. Menyediakan dashboard dan laporan manajemen.
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2. Stakeholder Analysis

Stakeholder	Peran
Yayasan	Sponsor

Kepala Sekolah	Decision Maker
Wakil Kepala Sekolah	Decision Maker
Admin Sekolah	Operational User
Guru	Operational User
Wali Kelas	Operational User
Siswa	Operational User
Orang Tua	External User
Staff Keuangan	Operational User
Panitia Event	Operational User
Auditor Internal	Support

3. Business Capability Map

Admission Domain

- PPDB Management
- Seleksi Siswa
- Registrasi Siswa
- Manajemen Calon Siswa

Student Management

- Student Profile
- Student Lifecycle
- Class Placement
- Student Promotion
- Alumni Management

Parent Management

- Parent Profile
- Parent Communication
- Parent Portal

Academic Management

- Curriculum Management
- Subject Management
- Class Management
- Schedule Management
- Attendance Management
- Assessment Management
- Report Card Management

Learning Outcome Assessment

- Outcome Definition
- Observation Management
- Evidence Collection
- Outcome Evaluation
- LOA Reporting

LMS

- Learning Material Management
- Assignment Submission
- Assignment Review

Event Management

- Event Planning
- Event Registration
- Event Execution
- Event Evaluation

Finance Management

- Registration Billing
- Tuition Billing
- Payment Collection
- Payment Verification

Human Resource

- Teacher Management
- Teacher Assignment
- Teacher Evaluation

Reporting & Analytics

- Executive Dashboard
 - Academic Dashboard
 - Financial Dashboard
 - Event Dashboard
 - LOA Dashboard
-

LEVEL 2 — BUSINESS STRUCTURE

Business Resource Catalog

Master Resource

- Student
- Parent
- Teacher
- Class
- Academic Year
- Subject
- Learning Outcome
- Event
- User
- Role

Transaction Resource

- Applicant
 - Admission Form
 - Registration
 - Attendance
 - Assignment
 - Assignment Submission
 - Academic Assessment
 - LOA Assessment
 - Event Registration
 - Billing
 - Payment
-

Activity Catalog

PPDB

- Create Applicant
- Verify Applicant
- Schedule Interview
- Conduct Assessment
- Approve Applicant
- Reject Applicant
- Register Student

Academic

- Create Schedule
- Record Attendance
- Create Assignment
- Submit Assignment
- Assess Assignment
- Generate Report Card

LOA

- Create Observation
- Upload Evidence
- Assess Outcome
- Publish Outcome Report

Event

- Create Event
- Open Registration
- Register Participant
- Execute Event
- Publish Evaluation

Finance

- Generate Bill
 - Verify Payment
 - Approve Payment
 - Print Receipt
-

ERP Navigation Blueprint

Dashboard

Master Data

- Student
- Parent
- Teacher
- Academic Year
- Class
- Subject
- Learning Outcome

Admissions

- Applicant
- Selection
- Registration

Academic

- Schedule
- Attendance
- Assignment
- Assessment
- Report Card

LOA

- Observation
- Evidence
- Evaluation
- Report

LMS

- Learning Material
- Assignment Submission

Events

- Event Calendar
- Event Registration
- Event Evaluation

Finance

- Billing
- Payment

Reports

- Academic Reports
- LOA Reports
- Finance Reports
- Event Reports

System

- User
 - Role
 - Audit Trail
-

LEVEL 3 — WORKFLOW DISCOVERY

Cross Domain Workflow

PPDB

- Registrasi
 - Menjadi Siswa
 - Penempatan Kelas
 - Aktivitas Akademik
 - LOA
 - Event Sekolah
 - Rapor
 - Kenaikan Kelas
 - Kelulusan
 - Alumni
-

Workflow 01 – Admission

Create Applicant

- Verify Applicant

- Interview
 - Assessment
 - Acceptance Decision
 - Registration
 - Student Created
-

Workflow 02 – Academic Learning

- Create Schedule
 - Teaching Session
 - Attendance
 - Assignment
 - Submission
 - Assessment
 - Report Card
-

Workflow 03 – LOA

- Observation
 - Evidence Collection
 - Outcome Assessment
 - Validation
 - Publish LOA Report
-

Workflow 04 – Event

- Create Event
 - Publish Event
 - Registration
 - Attendance
 - Evaluation
 - Publish Report
-

Workflow 05 – Tuition Payment

Generate Billing
→ Parent Payment
→ Verification
→ Receipt
→ Paid Status

Workflow Resource Matrix

Workflow	Input	Output
Admission	Applicant	Student
Academic	Student	Assessment
LOA	Observation	Outcome Report
Event	Event	Event Report
Billing	Student	Payment

LEVEL 4 — EVENT DISCOVERY & RULE MODELING

Event Catalog

Admission Event

- APPLICANT_CREATED
- APPLICANT_VERIFIED
- APPLICANT_APPROVED
- APPLICANT_REJECTED
- STUDENT_REGISTERED

Academic Event

- ATTENDANCE_RECORDED
- ASSIGNMENT_CREATED

- ASSIGNMENT_SUBMITTED
- ASSESSMENT_COMPLETED
- REPORT_CARD_PUBLISHED

LOA Event

- OBSERVATION_CREATED
- EVIDENCE_UPLOADED
- OUTCOME_ASSESSED
- LOA_PUBLISHED

Event Event

- EVENT_CREATED
- EVENT_PUBLISHED
- EVENT_REGISTERED
- EVENT_COMPLETED

Finance Event

- BILL_GENERATED
- PAYMENT_RECEIVED
- PAYMENT_VERIFIED

State Transition

Applicant

Draft

→ Submitted

→ Verified

→ Approved

→ Registered

Student

Registered

→ Active

→ Promoted

→ Graduated

→ Alumni

Billing

Draft

→ Issued

→ Paid

Business Rule Catalog

ADM-001

Calon siswa wajib melengkapi dokumen.

ADM-002

Siswa hanya dapat diregistrasi setelah diterima.

ACD-001

Tugas hanya dapat dinilai setelah dikumpulkan.

LOA-001

LOA wajib memiliki evidence.

FIN-001

Tagihan dianggap lunas setelah pembayaran diverifikasi.

LEVEL 5 — DATA MODELING

Core Entity

Master

- students
- parents
- teachers
- classes
- subjects
- academic_years
- learning_outcomes
- events

Transaction

- applicants

- registrations
- attendances
- assignments
- assignment_submissions
- academic_assessments
- loa_observations
- loa_evidences
- loa_assessments
- billings
- payments

System

- users
 - roles
 - audit_logs
 - event_logs
-

Conceptual ERD

Parent

1:N

Student

Class

1:N

Student

Teacher

1:N

Class

Student

1:N

Attendance

Student

1:N

Assignment Submission

Student

1:N

Academic Assessment

Student
1:N
LOA Assessment

Student
1:N
Billing

Billing
1:N
Payment

Event
1:N
Event Registration

Student
1:N
Event Registration

Audit Persistence

audit_logs

- id
 - table_name
 - record_id
 - action
 - before_value
 - after_value
 - user_id
 - created_at
-

Event Persistence

event_logs

- id
- event_code
- resource_type

- resource_id
 - actor_id
 - event_time
-

LEVEL 6 — APPLICATION ARCHITECTURE

Dashboard Architecture

Executive Dashboard

- Total Students
- New Admissions
- Active Classes
- Revenue Summary
- Upcoming Events

Academic Dashboard

- Attendance Rate
- Assignment Completion
- Assessment Summary

LOA Dashboard

- Outcome Achievement
- Observation Count
- Evidence Completion

Finance Dashboard

- Outstanding Bills
- Paid Bills
- Collection Trend

Event Dashboard

- Upcoming Events
- Participant Count
- Event Evaluation Score

User Flow

Parent

Login

- View Child Profile
- View Billing
- Pay Billing
- View Report Card
- View LOA Report
- Register Event

Teacher

Login

- Attendance
- Assignment
- Assessment
- Observation
- LOA Evaluation

Admin

Login

- Admission
 - Student Registration
 - Billing
 - Event Setup
-

Reporting Catalog

Academic

- Student Report Card
- Attendance Report
- Assessment Report

LOA

- Student LOA Report
- Outcome Achievement Report

Finance

- Billing Report
- Payment Report

Event

- Participant Report
- Event Evaluation Report

Management

- Student Statistics
 - Enrollment Trend
 - Revenue Trend
-

Monitoring Catalog

- Unverified Applicant
 - Overdue Billing
 - Missing Assessment
 - Missing LOA Evidence
 - Event Registration Low
 - Student Attendance Below Threshold
-

Audit Trail Catalog

Login Audit

- Login
- Logout

Activity Audit

- Create
- Update
- Delete
- Approve

Workflow Audit

- Registration
- Assessment
- Billing
- Payment
- Event Participation

Data Change Audit

- Before Value
- After Value

END OF DOCUMENT

Revisi Level 1

Business Goal Tambahan

Goal 9

Menyediakan sistem dokumentasi pembelajaran harian yang terstruktur.

Goal 10

Mengubah seluruh aktivitas pembelajaran menjadi evidence pembelajaran.

Goal 11

Menghasilkan portofolio siswa secara otomatis.

Goal 12

Menghasilkan rapor semester berbasis evidence dan outcome.

Revisi Capability Map

Tambahkan Domain Baru

Learning Planning Domain

Capability:

- Semester Planning
 - Weekly Planning
 - Daily Planning
 - Learning Theme Management
 - Learning Outcome Mapping
 - Learning Activity Design
-

Daily Learning Domain

Capability:

- Daily Attendance
 - Daily Activity Recording
 - Daily Observation
 - Daily Assessment
 - Daily Reporting
-

Student Portfolio Domain

Capability:

- Portfolio Collection
 - Portfolio Compilation
 - Portfolio Publishing
-

Student Progress Domain

Capability:

- Progress Monitoring
- Progress Reporting
- Parent Communication

Revisi Level 2

Resource Catalog Baru

Planning Resource

- Semester Plan
- Weekly Plan
- Daily Plan
- Theme
- Learning Objective
- Learning Outcome

Execution Resource

- Daily Activity
- Attendance
- Observation
- Daily Assessment
- Daily Report

Evidence Resource

- Learning Evidence
- Photo Evidence
- Video Evidence
- Student Work

Outcome Resource

- Portfolio Item
- Portfolio
- Progress Report
- Semester Report

Revisi Level 3

Workflow Inti ERP

Workflow 1

Academic Planning

Semester Plan
→ Weekly Plan
→ Daily Plan

Workflow 2

Learning Execution

Daily Plan
→ Learning Activity
→ Attendance
→ Observation
→ Assessment

Workflow 3

Daily Communication

Observation
→ Daily Assessment
→ Daily Report
→ Parent Portal

Workflow 4

Portfolio Building

Daily Activity
→ Evidence
→ Portfolio Item
→ Portfolio

Workflow 5

Progress Monitoring

Daily Assessment

→ Weekly Summary

→ Monthly Summary

→ Progress Report

Workflow 6

Semester Reporting

Daily Assessment

+

Portfolio

+

Learning Outcome Achievement

→ Semester Report

→ Rapor

Revisi Level 4

Event Baru

Planning Event

- SEMESTER_PLAN_CREATED
- WEEKLY_PLAN_CREATED
- DAILY_PLAN_CREATED

Learning Event

- ACTIVITY_EXECUTED
- OBSERVATION_RECORDED
- DAILY_ASSESSMENT_COMPLETED

Reporting Event

- DAILY_REPORT_PUBLISHED
- PROGRESS_REPORT_PUBLISHED

- SEMESTER_REPORT_PUBLISHED

Portfolio Event

- EVIDENCE_UPLOADED
 - PORTFOLIO_ITEM_CREATED
-

Revisi Level 5

Resource Prioritas Tinggi

Sebelumnya:

Student
Teacher
Assessment

Menjadi:

Student
Facilitator
SemesterPlan
WeeklyPlan
DailyPlan
DailyActivity
Observation
Assessment
DailyReport
Evidence
PortfolioItem
Portfolio
ProgressReport
SemesterReport

Revisi Level 6

Menu Utama ERP

Dashboard

Learning Planning

- Semester Plan
- Weekly Plan
- Daily Plan

Learning Execution

- Attendance
- Daily Activity
- Observation
- Daily Assessment

Daily Report

- Create Report
- Publish Report

Portfolio

- Portfolio Item
- Student Portfolio

Progress

- Progress Report

Semester Report

- Semester Assessment
- Semester Report
- Rapor

Students

- Student Profile
- Parent Profile

Events

- Outbound
- Camping
- Outing Class
- Market Day
- Pentas Seni

- Wisuda
- Parenting Seminar

Finance

- Billing
 - Payment
-

Kesimpulan Arsitektural

Setelah melihat contoh Daily Report yang Anda unggah, saya justru menyimpulkan bahwa:

ERP Sekolah Alam bukan Student Information System (SIS).

ERP Sekolah Alam adalah Outcome-Based Learning Management System yang dibangun di atas siklus:

Plan

- Do
- Observe
- Assess
- Report
- Portfolio
- Progress
- Semester Report

Jika nanti Anda membuat full mockup, saya akan menyarankan bahwa layar yang paling banyak justru bukan:

- Nilai
- Mata pelajaran
- Jadwal

melainkan:

- Daily Plan
- Weekly Plan
- Daily Activity
- Observation
- Daily Assessment
- Daily Report
- Portfolio

- Progress Report
- Semester Report

karena artefak-artefak itulah yang sebenarnya menjadi jantung operasional Sekolah Alam dan sumber data utama rapor berbasis OBE.